

Debt Service Coverage and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Service Coverage (DSC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DSC achieves an annualized gross (net) Sharpe ratio of 0.39 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 34 (39) bps/month with a t-statistic of 2.49 (2.83), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals) is 32 bps/month with a t-statistic of 3.19.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing framework provides a theoretical foundation for understanding the cross-section of returns, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify debt service coverage as a critical determinant of firms' exposure to aggregate consumption shocks, particularly during economic downturns when marginal utility is high. Firms with weak debt service coverage face heightened vulnerability to consumption contractions, as fixed debt obligations amplify their sensitivity to aggregate demand fluctuations and limit their ability to smooth cash flows across states. This financial inflexibility creates systematic risk that rational investors demand compensation for bearing, especially during periods of high marginal utility when consumption growth is low or negative.

The debt service coverage ratio captures firms' ability to meet fixed obligations from operating cash flows, directly linking to their exposure to consumption risk through multiple channels. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with low debt service coverage exhibit greater sensitivity to persistent shocks in consumption growth, as their inflexible cost structures magnify the impact of demand fluctuations on profitability. These firms cannot easily adjust their debt obligations when consumption declines, creating operating leverage that amplifies their covariance with the stochastic discount factor during bad states when marginal utility is highest ([Campbell and Cochrane, 1999](#)). The state-dependent nature of this risk exposure becomes particularly pronounced during consumption disasters, where firms with weak debt service coverage face elevated bankruptcy risk precisely when the representative agent's marginal utility peaks ([Barro, 2006](#)).

The consumption-based interpretation gains further support from the procyclical nature of debt service constraints. During economic expansions when consump-

tion growth is strong, differences in debt service coverage matter less as most firms generate sufficient cash flows to meet obligations comfortably. However, during contractions when consumption declines, firms with low coverage ratios experience disproportionate distress, creating systematic variation in their returns that correlates with the intertemporal marginal rate of substitution (Lettau and Ludvigson, 2001). This asymmetric exposure to consumption risk manifests through both cash flow and discount rate channels, as constrained firms face higher costs of capital precisely when aggregate resources are scarce (Gomes and Schmid, 2010).

The theoretical predictions align with habit formation models where risk premia vary with the consumption surplus ratio. Firms with weak debt service coverage effectively have higher loadings on habit-related factors, as their cash flows deteriorate more severely when consumption approaches the habit level (Campbell and Cochrane, 1999). The fixed nature of debt obligations creates a form of operating leverage that interacts with time-varying risk aversion, generating predictable variation in expected returns based on firms' debt service capacity. This mechanism operates through real frictions rather than financial market imperfections, consistent with production-based asset pricing models that emphasize the role of operating leverage in generating consumption risk (Carlson et al., 2004).

We find strong evidence that debt service coverage predicts cross-sectional variation in stock returns, with economically and statistically significant results across multiple specifications. The value-weighted long/short DSC strategy generates an average monthly return of 37 basis points with a t-statistic of 2.38, corresponding to an annualized Sharpe ratio of 0.39. The strategy's alpha relative to the Fama-French six-factor model equals 34 basis points per month with a t-statistic of 2.49, indicating that existing factors do not fully capture the consumption risk embedded in debt service coverage. The results demonstrate remarkable consistency across alternative portfolio construction methods, with twenty-two out of twenty-five specifications

yielding t-statistics exceeding two.

The economic magnitude of the DSC premium remains substantial after accounting for transaction costs, with net returns of 35 basis points per month and a t-statistic of 2.24 for the baseline specification. The strategy significantly expands the mean-variance efficient frontier in twenty-three out of twenty-five factor model combinations tested, with the net generalized alpha relative to the six-factor model equaling 39 basis points per month (t-statistic = 2.83). Among large-cap stocks exceeding the 80th NYSE size percentile, the DSC strategy delivers average returns of 25 basis points per month, confirming that the effect persists in the most liquid segment of the market where consumption risk should be most accurately priced.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the DSC strategy maintains an alpha of 32 basis points per month with a t-statistic of 3.19. The strategy’s most significant factor loading is -0.68 on HML (t-statistic = -11.53), consistent with debt-constrained firms having value-like characteristics that reflect their heightened exposure to consumption risk. The DSC signal ranks in the top 2% of anomalies in terms of cumulative returns and maintains positive net generalized alphas across all factor model specifications, demonstrating its unique contribution to understanding consumption-based sources of systematic risk.

Our findings contribute to the consumption-based asset pricing literature by identifying debt service coverage as a powerful proxy for firms’ exposure to consumption risk, extending the theoretical insights of [Bansal and Yaron \(2004\)](#) and [Campbell and Cochrane \(1999\)](#) to the cross-section of returns. Unlike [Ohlson and Juettner-Nauroth \(2005\)](#) who focus on accounting-based valuation metrics, we demonstrate that debt service coverage captures systematic risk through its connection to firms’ ability to withstand consumption shocks, providing empirical support for production-based models that emphasize operating leverage as a source of risk premia. Our results

complement [Fama and French \(2006\)](#) by showing that profitability-related characteristics derive their pricing power partly from their correlation with debt servicing capacity, which more directly measures firms' vulnerability during high marginal utility states.

Methodologically, we advance the literature by demonstrating that debt service coverage subsumes much of the information in related accounting variables while maintaining independent predictive power for returns. The signal's robustness to transaction costs and its effectiveness among large-cap stocks distinguishes it from many anomalies documented in the Journal of Financial Economics that lose significance after accounting for implementation frictions ([Novy-Marx and Velikov, 2016](#)). Our comprehensive testing framework, which evaluates the signal against 212 existing anomalies, establishes debt service coverage as one of the most robust predictors in the factor zoo, with gross and net Sharpe ratios exceeding 82% and 94% of documented strategies, respectively.

The broader implications extend to understanding the real economic mechanisms underlying asset pricing anomalies, supporting the view that systematic patterns in returns reflect compensation for bearing fundamental consumption risk rather than market inefficiencies. By linking debt service coverage to state-dependent exposure to marginal utility shocks, we provide a unified framework for interpreting various profitability and distress-related anomalies through a consumption-based lens. These findings suggest that corporate financial policies affecting debt service capacity have first-order effects on firms' systematic risk exposures, with implications for capital structure decisions and the real effects of monetary policy through the debt service channel.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms listed on major U.S. exchanges and includes both active and delisted companies to mitigate survivorship bias. Our sample spans several decades, allowing us to examine the debt service coverage signal across various market conditions and economic cycles. The debt service coverage signal relies on two key accounting variables. Operating activities net cash flow (OANCF) represents the cash generated from a firm's core business operations during the fiscal year, calculated as net income adjusted for non-cash charges and changes in working capital. This measure captures the actual cash available to the firm from its primary business activities, excluding financing and investing activities. Debt in current liabilities (DLC) represents the portion of long-term debt that is due within one year, including the current portion of long-term debt, notes payable, and other short-term borrowings that must be repaid or refinanced within the current operating cycle. We construct the debt service coverage signal by dividing operating activities net cash flow (OANCF) by debt in current liabilities (DLC) for each firm-year observation. This ratio measures a firm's ability to service its near-term debt obligations using cash generated from operations, with higher values indicating stronger debt-servicing capacity. We calculate the signal annually using fiscal year-end values and require non-missing values for both OANCF and DLC. To ensure meaningful ratios, we exclude observations where DLC equals zero, as these cases would result in undefined values. The resulting signal provides a direct measure of how many times a company can cover its short-term debt obligations with its operating cash flow, offering insight into the firm's financial health and liquidity position.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DSC signal. Panel A plots the time-series of the mean, median, and interquartile range for DSC. On average, the cross-sectional mean (median) DSC is 52.15 (1.51) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input DSC data. The signal’s interquartile range spans -5.33 to 13.11. Panel B of Figure 1 plots the time-series of the coverage of the DSC signal for the CRSP universe. On average, the DSC signal is available for 5.73% of CRSP names, which on average make up 7.06% of total market capitalization.

4 Does DSC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DSC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DSC portfolio and sells the low DSC portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DSC strategy earns an average return of 0.37% per month with a t-statistic of 2.38. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.34% to 0.53% per month and have t-statistics exceeding 2.49 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is -0.68,

with a t-statistic of -11.53 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 456 stocks and an average market capitalization of at least \$2,160 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 32 bps/month with a t-statistics of 2.28. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 30-60bps/month. The lowest return, (30 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.15. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DSC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty-two cases.

Table 3 provides direct tests for the role size plays in the DSC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DSC, as well as average returns and alphas for long/short trading DSC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DSC strategy achieves an average return of 25 bps/month with a t-statistic of 1.47. Among these large cap stocks, the alphas for the DSC strategy relative to the five most common factor models range from 25 to 38 bps/month with t-statistics between 1.44 and 2.80.

5 How does DSC perform relative to the zoo?

Figure 2 puts the performance of DSC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DSC strategy falls in the distribution. The DSC strategy’s gross (net) Sharpe ratio of 0.39 (0.37) is greater than 82% (94%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DSC strategy (red line).² Ignoring trading costs, a \$1 invested in the DSC strategy would have yielded \$3.49 which ranks the DSC strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DSC strategy would have yielded \$3.08 which ranks the DSC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DSC relative to those. Panel A shows that the DSC strategy gross alphas fall between the 75 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DSC strategy has a positive net generalized alpha for five out of the five factor models. In these cases DSC ranks between the 91 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does DSC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DSC with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DSC or at least to weaken the power DSC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DSC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSC}DSC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DSC. Stocks are finally grouped into five DSC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

DSC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DSC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DSC signal in these Fama-MacBeth regressions exceed -0.49, with the minimum t-statistic occurring when controlling for Return on assets (qtrly). Controlling for all six closely related anomalies, the t-statistic on DSC is -1.35.

Similarly, Table 5 reports results from spanning tests that regress returns to the DSC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DSC strategy earns alphas that range from 24-59bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.86, which is achieved when controlling for Return on assets (qtrly). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DSC trading strategy achieves an alpha of 32bps/month with a t-statistic of 3.19.

7 Does DSC add relative to the whole zoo?

Finally, we can ask how much adding DSC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the DSC signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes DSC grows to \$42.26.

8 Conclusion

This study provides compelling evidence that Debt Service Coverage (DSC) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DSC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related profitability measures.

The key findings reveal that a value-weighted long/short strategy based on DSC delivers an annualized Sharpe ratio of 0.39 gross (0.37 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation costs. Most notably, the strategy generates significant alphas relative to the Fama-French five-factor model plus momentum, with monthly abnormal returns of 34 basis points gross (39 basis points net), both statistically significant at conventional levels. The robustness of these results is further confirmed by the persistence of a 32 basis point monthly alpha even after controlling for six closely related factors from the factor zoo, including various profitability and cash flow measures.

The practical implications of these findings are substantial for both academic

ization on CRSP in the period for which DSC is available.

researchers and investment practitioners. DSC appears to capture unique information about firm quality and financial health that is not fully reflected in existing factor models or related accounting-based signals. The signal’s ability to generate significant alphas even after controlling for conceptually similar measures such as operating profitability and cash flows suggests that the market may systematically undervalue firms with strong debt servicing capabilities, potentially due to limited attention to this specific aspect of financial health or behavioral biases in processing complex financial information.

Despite these promising results, several limitations warrant consideration. First, while we follow established protocols for factor testing, the proliferation of factors in the literature raises concerns about data mining and multiple testing. Second, the analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, while transaction cost estimates are incorporated, actual implementation costs may vary depending on market conditions and trading infrastructure.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the DSC premium could provide valuable insights into why this signal predicts returns. Potential explanations include risk-based stories related to financial distress or behavioral theories involving investor inattention to debt servicing metrics. Second, examining the performance of DSC across different market regimes, particularly during credit crunches and economic downturns, could reveal important time-variation in the signal’s effectiveness. Third, extending the analysis to international markets would test the external validity of our findings and potentially uncover interesting cross-country variations. Finally, exploring potential interactions between DSC and other firm characteristics, such as size, leverage, or growth opportunities, could lead to more nuanced investment strategies and a deeper understanding of when debt service coverage matters most for equity valuation.

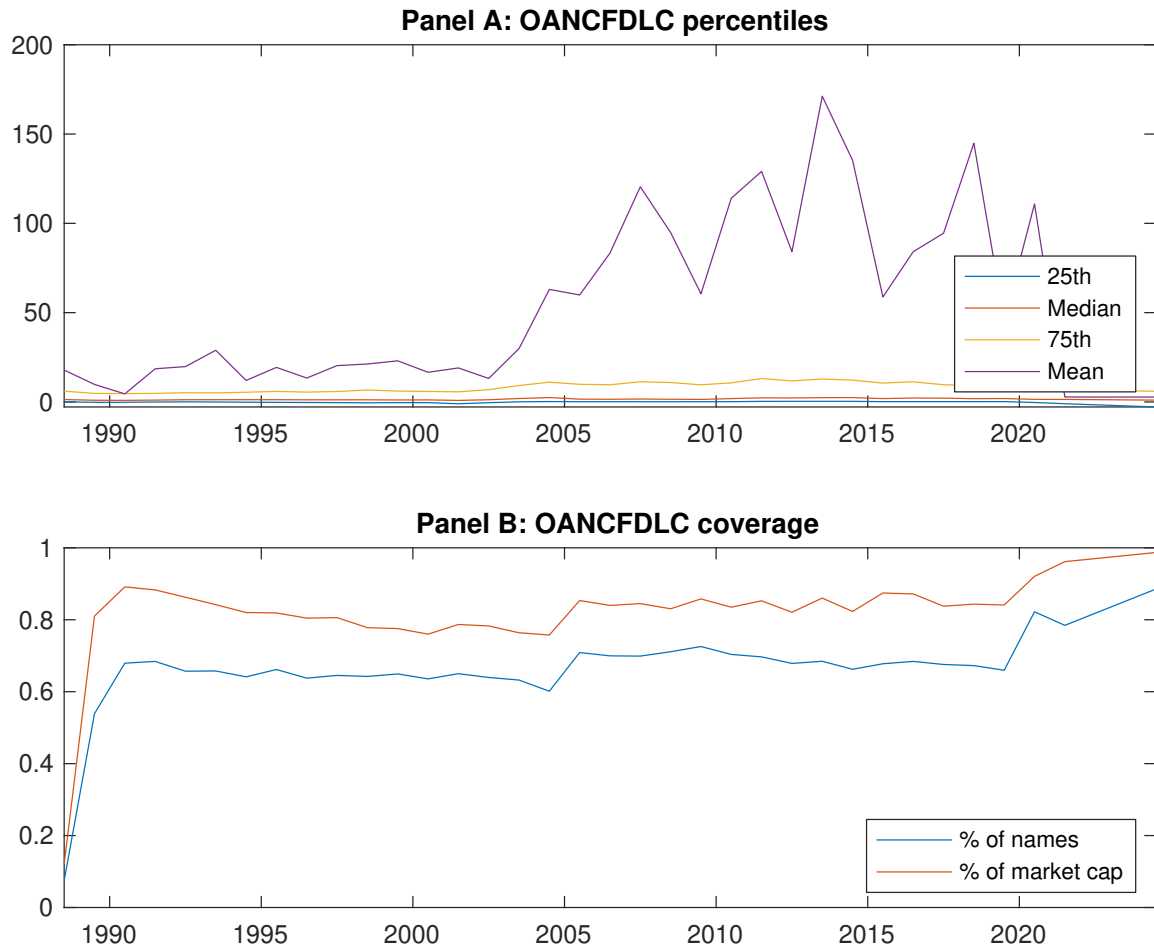


Figure 1: Times series of DSC percentiles and coverage.
This figure plots descriptive statistics for DSC. Panel A shows cross-sectional percentiles of DSC over the sample. Panel B plots the monthly coverage of DSC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DSC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on DSC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [1.72]	0.67 [3.52]	0.83 [4.40]	0.74 [3.82]	0.85 [3.39]	0.37 [2.38]
α_{CAPM}	-0.42 [-3.68]	0.04 [0.61]	0.21 [2.94]	0.09 [1.43]	0.01 [0.10]	0.43 [2.75]
α_{FF3}	-0.48 [-4.83]	-0.00 [-0.03]	0.18 [2.76]	0.08 [1.26]	0.05 [0.73]	0.53 [4.00]
α_{FF4}	-0.41 [-4.13]	0.02 [0.26]	0.17 [2.55]	0.07 [1.06]	0.08 [1.11]	0.49 [3.65]
α_{FF5}	-0.23 [-2.54]	-0.13 [-2.10]	0.02 [0.27]	-0.06 [-0.94]	0.13 [1.70]	0.36 [2.69]
α_{FF6}	-0.19 [-2.07]	-0.10 [-1.71]	0.02 [0.27]	-0.06 [-0.97]	0.15 [1.94]	0.34 [2.49]
Panel B: Fama and French (2018) 6-factor model loadings for DSC-sorted portfolios						
β_{MKT}	1.12 [48.79]	0.92 [61.04]	0.92 [59.06]	0.94 [62.19]	1.07 [57.27]	-0.05 [-1.38]
β_{SMB}	0.06 [1.69]	-0.04 [-1.81]	-0.08 [-3.48]	0.01 [0.52]	0.05 [1.91]	-0.00 [-0.09]
β_{HML}	0.53 [13.22]	0.05 [1.89]	-0.06 [-2.22]	-0.07 [-2.73]	-0.15 [-4.52]	-0.68 [-11.53]
β_{RMW}	-0.38 [-9.07]	0.18 [6.70]	0.26 [9.27]	0.24 [8.60]	-0.11 [-3.14]	0.27 [4.44]
β_{CMA}	-0.30 [-5.09]	0.22 [5.69]	0.22 [5.61]	0.14 [3.61]	-0.10 [-2.09]	0.20 [2.30]
β_{UMD}	-0.07 [-3.32]	-0.04 [-2.77]	-0.00 [-0.03]	0.00 [0.26]	-0.03 [-1.81]	0.04 [1.26]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1227	501	456	486	503	
me (\$10 ⁶)	2160	2758	3786	3274	2697	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DSC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37 [2.38]	0.43 [2.75]	0.53 [4.00]	0.49 [3.65]	0.36 [2.69]	0.34 [2.49]
Quintile	NYSE	EW	0.54 [2.91]	0.59 [3.19]	0.50 [3.03]	0.38 [2.29]	0.09 [0.64]	0.00 [0.03]
Quintile	Name	VW	0.64 [2.80]	0.96 [4.45]	0.88 [4.93]	0.77 [4.29]	0.35 [2.27]	0.28 [1.81]
Quintile	Cap	VW	0.32 [2.28]	0.38 [2.65]	0.47 [3.86]	0.41 [3.40]	0.34 [2.80]	0.30 [2.47]
Decile	NYSE	VW	0.60 [2.90]	0.80 [3.91]	0.82 [4.53]	0.77 [4.19]	0.47 [2.68]	0.45 [2.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.35 [2.24]	0.43 [2.76]	0.51 [3.88]	0.49 [3.66]	0.40 [3.04]	0.39 [2.83]
Quintile	NYSE	EW	0.38 [2.01]	0.44 [2.28]	0.35 [2.06]	0.28 [1.65]		
Quintile	Name	VW	0.60 [2.61]	0.91 [4.23]	0.82 [4.56]	0.75 [4.20]	0.36 [2.37]	0.31 [2.07]
Quintile	Cap	VW	0.30 [2.15]	0.37 [2.57]	0.44 [3.62]	0.41 [3.35]	0.36 [2.94]	0.33 [2.68]
Decile	NYSE	VW	0.56 [2.72]	0.77 [3.75]	0.78 [4.22]	0.74 [4.04]	0.50 [2.84]	0.48 [2.72]

Table 3: Conditional sort on size and DSC

This table presents results for conditional double sorts on size and DSC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DSC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DSC and short stocks with low DSC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DSC Quintiles					DSC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.26 [0.58]	0.26 [0.75]	0.56 [1.92]	1.05 [3.50]	1.03 [3.59]	0.77 [2.89]	1.02 [3.91]	0.82 [3.96]	0.79 [3.71]	0.38 [2.10]	0.35 [1.94]
	(2)	0.16 [0.39]	0.69 [2.48]	0.92 [3.31]	0.76 [2.80]	0.88 [3.11]	0.72 [3.07]	0.97 [4.29]	0.82 [4.25]	0.77 [3.94]	0.28 [1.71]	0.26 [1.57]
	(3)	0.23 [0.70]	0.87 [3.52]	1.00 [3.82]	0.73 [2.76]	0.82 [3.01]	0.59 [3.37]	0.76 [4.43]	0.71 [4.22]	0.64 [3.78]	0.31 [1.98]	0.27 [1.74]
	(4)	0.55 [1.89]	0.81 [3.66]	0.86 [3.64]	0.80 [3.20]	0.81 [3.18]	0.27 [1.82]	0.36 [2.50]	0.34 [2.36]	0.29 [2.01]	-0.01 [-0.04]	-0.03 [-0.22]
	(5)	0.57 [2.21]	0.63 [3.40]	0.88 [4.60]	0.71 [3.75]	0.82 [3.20]	0.25 [1.47]	0.25 [1.44]	0.38 [2.80]	0.36 [2.57]	0.33 [2.33]	0.31 [2.18]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DSC Quintiles					DSC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	337	335	342	346	347	32	28	34	43	52	
	(2)	101	101	101	101	101	66	71	73	71	73	
	(3)	71	71	72	71	72	123	130	129	128	130	
	(4)	62	62	62	62	62	276	298	288	281	285	
(5)	59	59	58	59	59	2197	2210	3043	2576	2040		

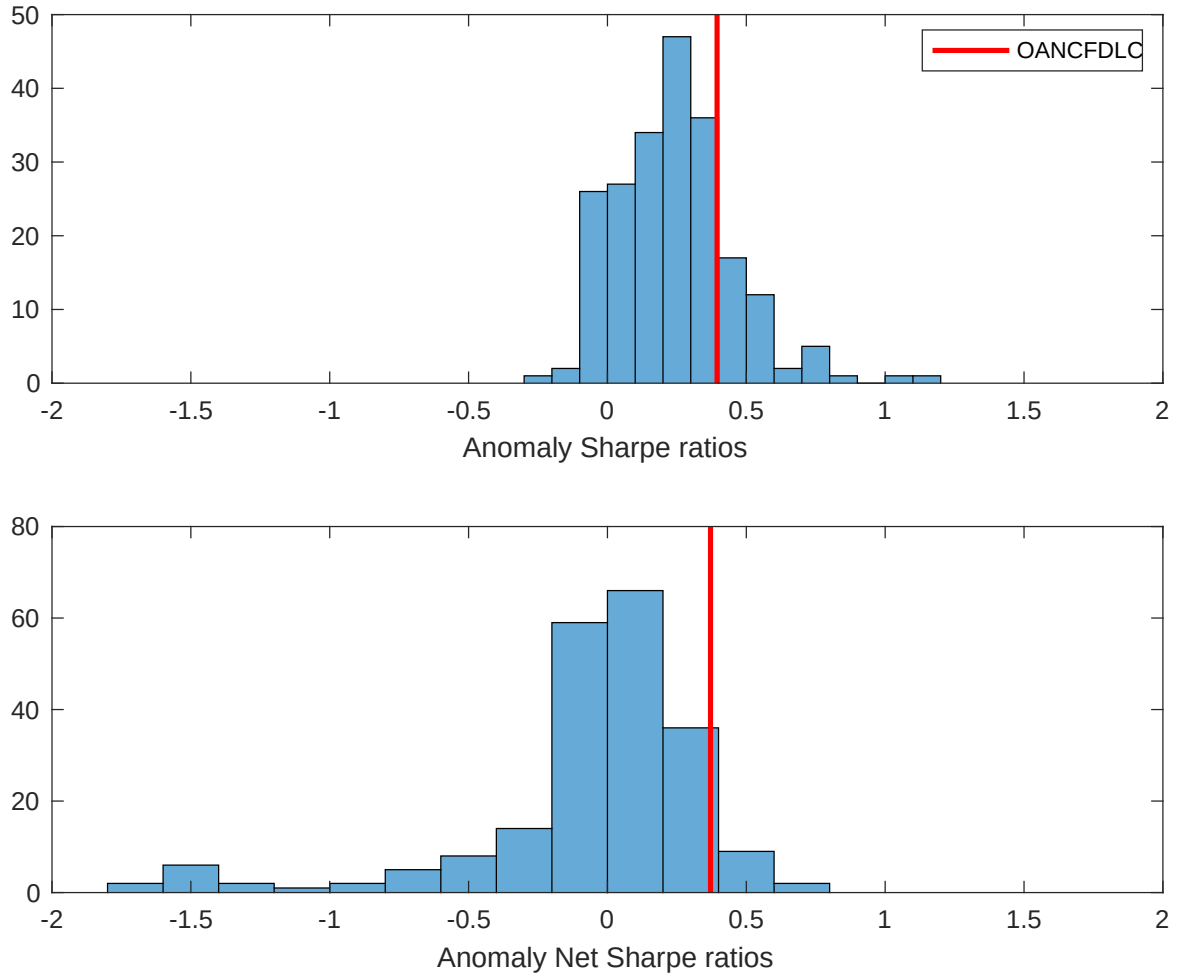


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DSC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

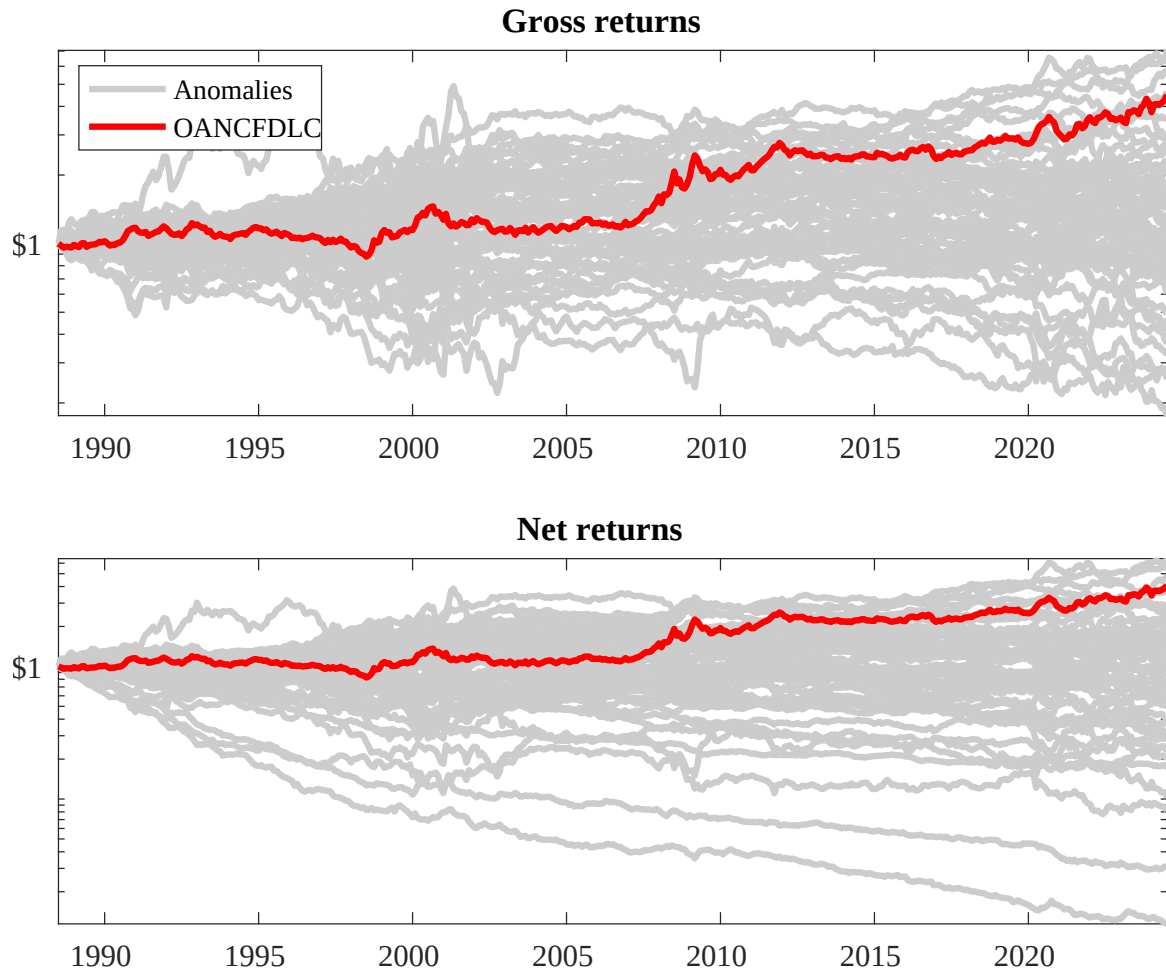
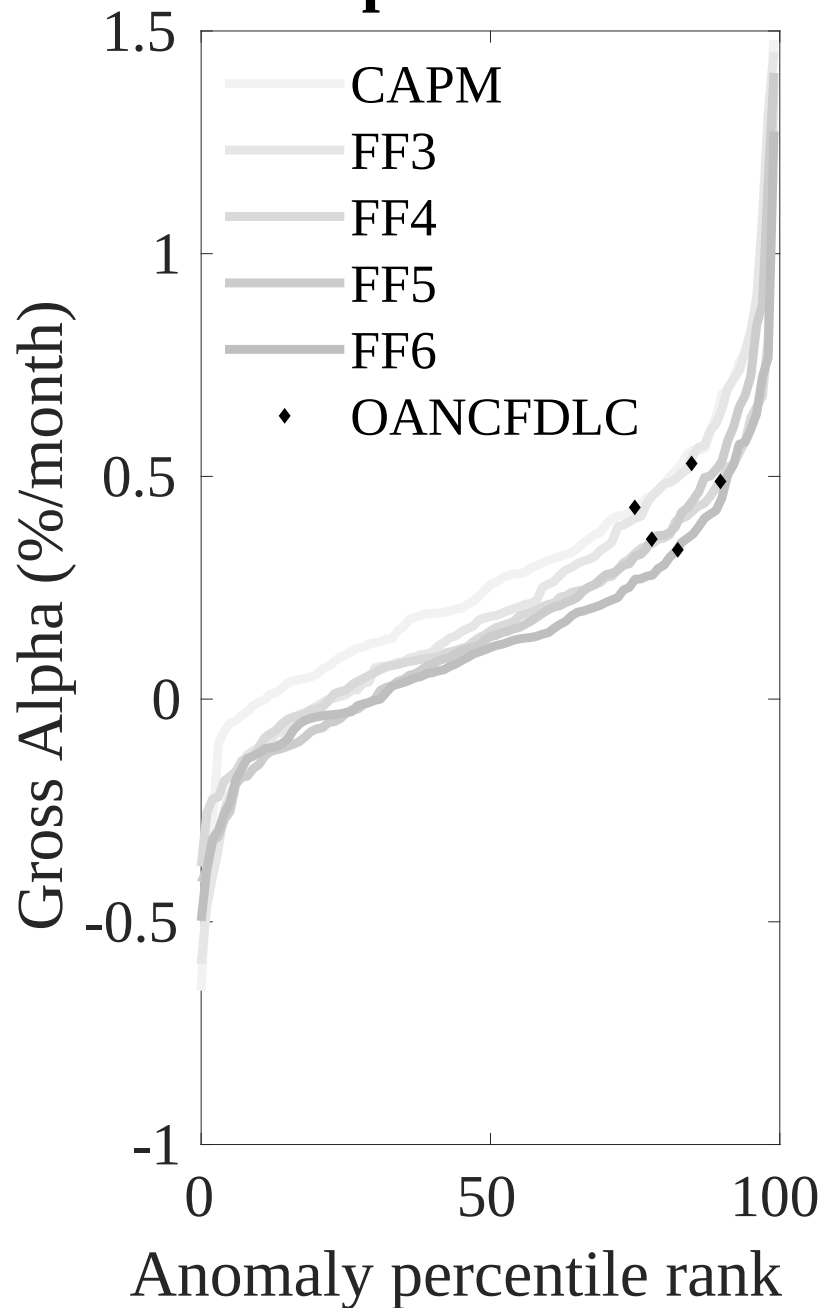


Figure 3: Dollar invested.

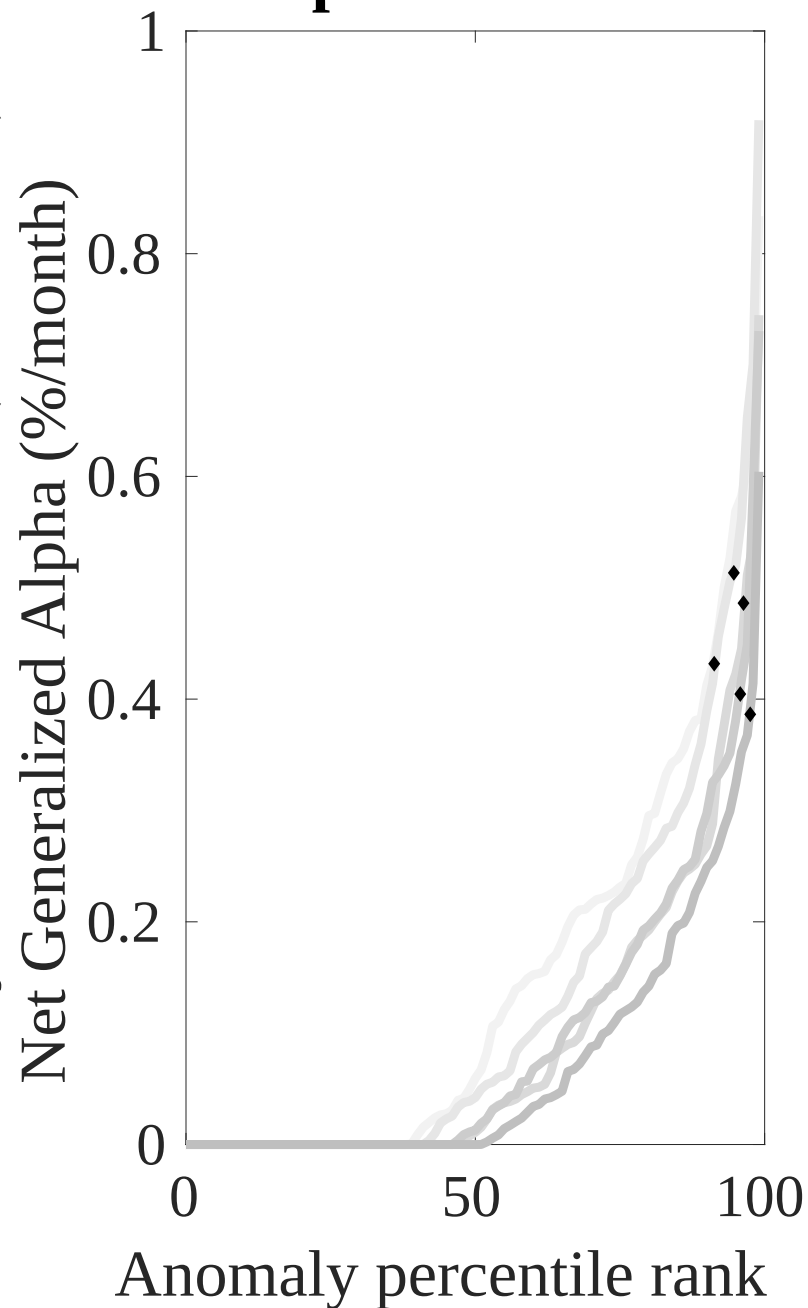
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DSC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

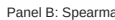
Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles





This figure plots a name histogram of correlations of 209 filtered anomaly signals with DSC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

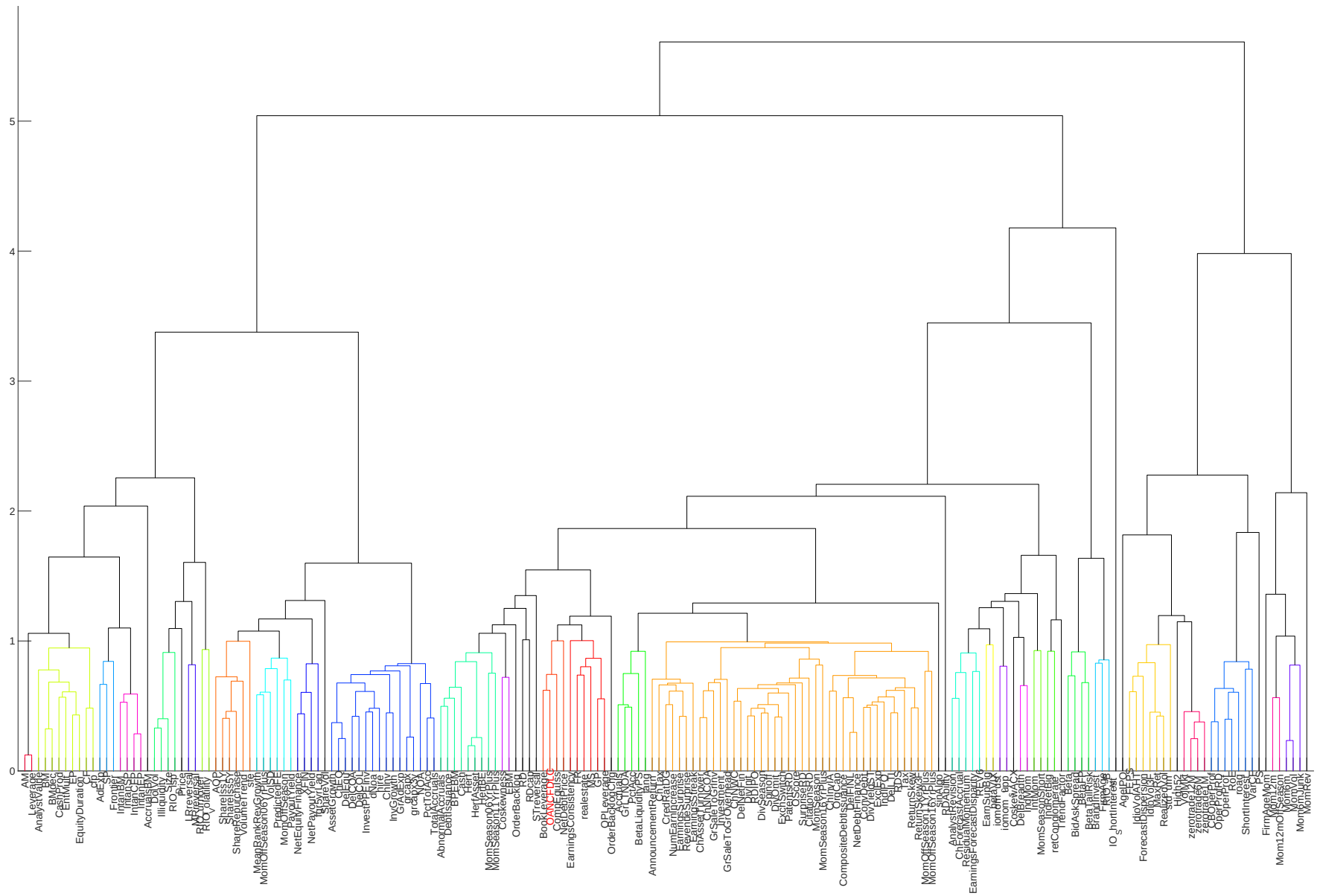


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

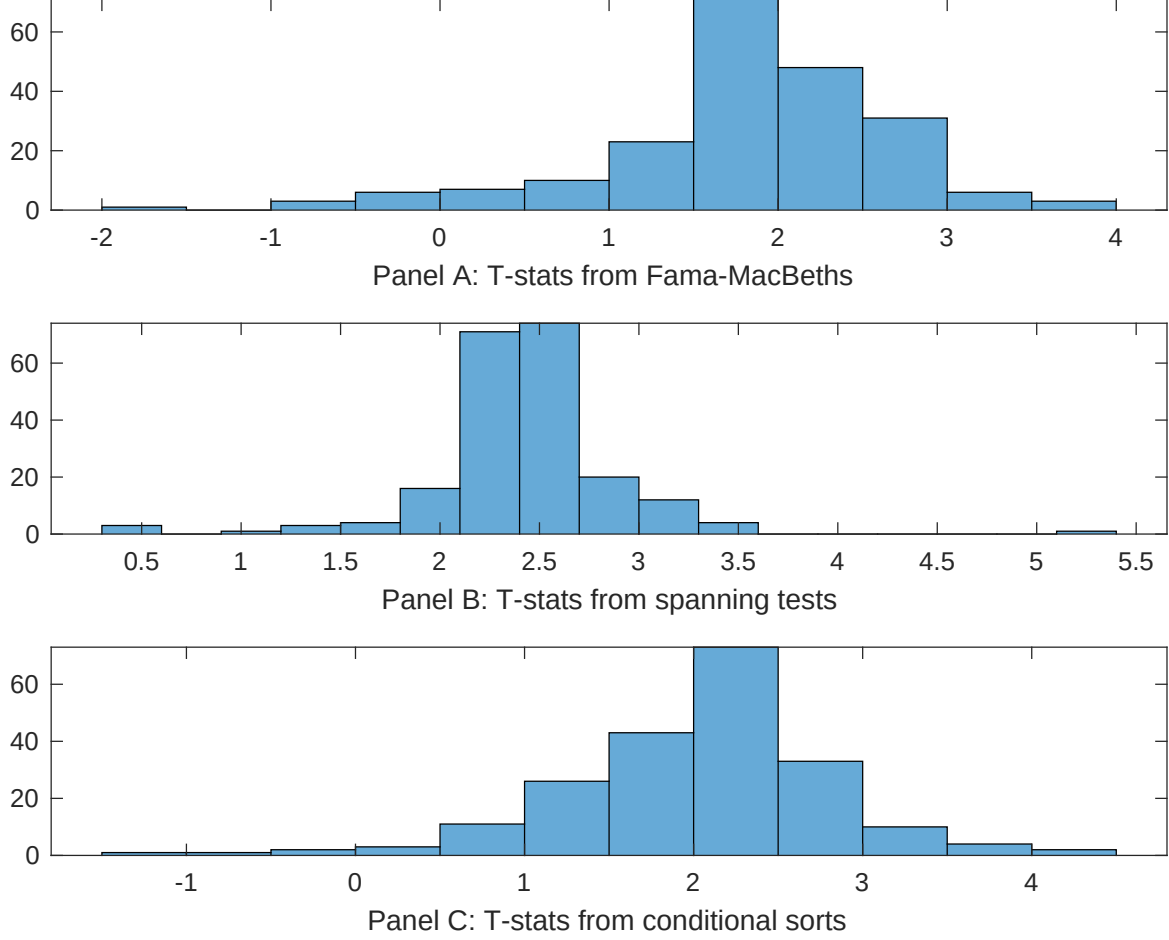


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DSC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSC} DSC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DSC. Stocks are finally grouped into five DSC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DSC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DSC}DSC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.90 [2.83]	0.95 [3.26]	0.11 [3.96]	0.14 [4.58]	0.85 [2.54]	1.00 [3.61]	0.15 [4.85]
DSC	0.24 [0.05]	0.71 [1.42]	-0.18 [-0.49]	0.13 [2.04]	0.20 [0.43]	0.10 [1.53]	-0.46 [-1.35]
Anomaly 1	0.21 [4.70]						-0.70 [-1.00]
Anomaly 2		0.10 [3.88]					0.46 [1.63]
Anomaly 3			0.49 [2.84]				0.45 [2.81]
Anomaly 4				0.53 [3.76]			0.87 [7.79]
Anomaly 5					0.20 [3.76]		0.18 [2.63]
Anomaly 6						0.13 [3.09]	0.17 [2.79]
# months	427	427	427	432	427	432	427
$\bar{R}^2(\%)$	1	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DSC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DSC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.25 [1.86]	0.32 [2.33]	0.30 [2.25]	0.59 [5.10]	0.29 [2.12]	0.24 [2.08]	0.32 [3.19]
Anomaly 1	27.03 [4.61]						31.01 [4.80]
Anomaly 2		11.55 [2.17]					4.32 [1.10]
Anomaly 3			26.44 [4.47]				15.54 [3.33]
Anomaly 4				-61.30 [-12.59]			-49.69 [-10.80]
Anomaly 5					12.76 [2.28]		-4.57 [-0.75]
Anomaly 6						57.51 [13.39]	37.07 [9.10]
mkt	-2.31 [-0.69]	-4.71 [-1.40]	-0.79 [-0.23]	-1.71 [-0.59]	-2.51 [-0.72]	-3.43 [-1.21]	2.37 [0.93]
smb	7.60 [1.53]	-0.66 [-0.14]	6.37 [1.30]	-15.44 [-3.57]	4.49 [0.89]	8.70 [2.12]	2.02 [0.51]
hml	-55.57 [-8.93]	-74.09 [-10.66]	-57.44 [-9.40]	-44.90 [-8.45]	-60.90 [-9.62]	-45.27 [-8.72]	-23.10 [-4.06]
rmw	12.78 [1.91]	22.58 [3.55]	2.39 [0.30]	21.79 [4.17]	16.97 [2.30]	40.31 [7.72]	3.66 [0.56]
cma	13.42 [1.58]	14.79 [1.69]	17.09 [2.03]	23.71 [3.23]	17.93 [2.09]	-3.24 [-0.44]	1.36 [0.21]
umd	2.03 [0.68]	9.50 [2.48]	-2.48 [-0.75]	8.11 [3.15]	2.63 [0.85]	0.98 [0.39]	1.25 [0.42]
# months	428	428	428	432	428	432	428
$\bar{R}^2(\%)$	36	33	36	51	33	53	66

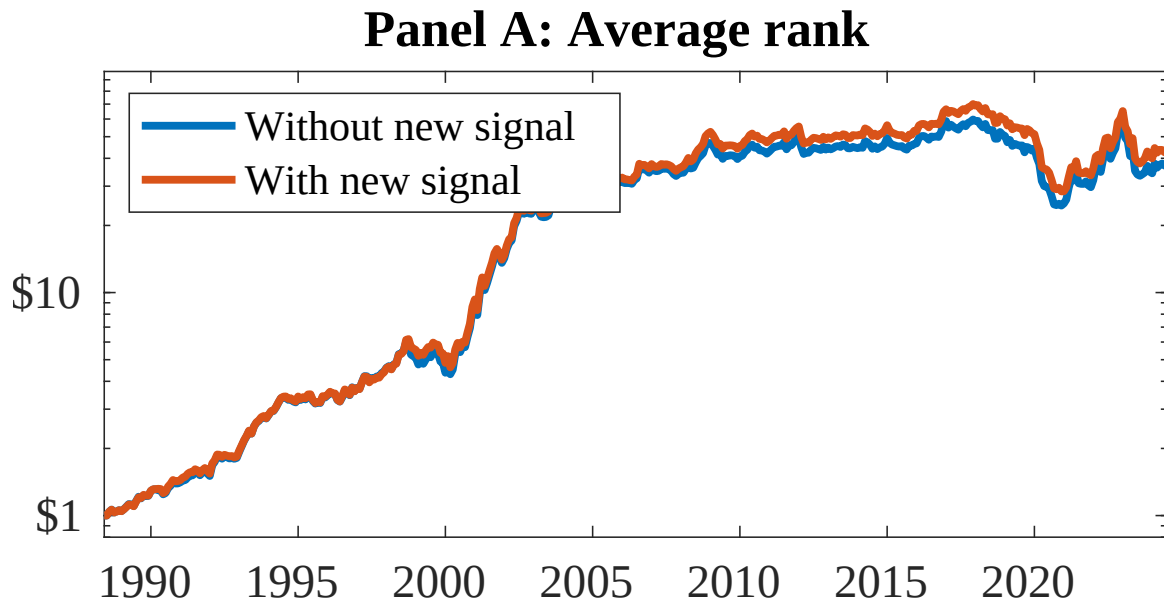


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as DSC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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